Data Analytics Project: Car Data Dataset

Submited By:

Team 4

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Dataset Used: cardata.csv

This document outlines a structured, end-to-end data analysis project based on game reviews from the car data dataset. The project is divided into ten scenarios, increasing in difficulty from basic data handling to advanced analysis and visualization, ensuring comprehensive skill coverage.

Project Goal: To analyse game review scores, platform popularity, and genre distribution over time using the cardata.csv dataset.

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (np.diff).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

🎯 Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling (Mean/Mode), Type Conversion, Column Removal
Pandas Operations	Grouping, Filtering, Aggregation, Feature Engineering (score_category, editors_choice conversion)
NumPy Calculations	Array conversion, Yearly score growth calculation (np.diff)
Visualization	Line Chart, Bar Chart, Pie Chart, Histogram, Stacked Bar Chart, Saving Plots

Scenario 1: Data Loading & Basic Cleaning (🟢 Easy)

Description:

In this scenario, the car dataset is loaded using Pandas to understand its structure and contents. Initial exploration is performed by displaying the first and last rows, column names, shape, and data types. Missing values are identified in important columns such as Selling Price, Present Price, Kms Driven, and Fuel Type. These missing values are handled by filling numerical columns with mean values and categorical columns with mode. Data type conversion is applied to ensure numerical columns are in the correct format. NumPy is then used to convert columns into arrays and perform statistical calculations such as minimum, maximum, and average selling price. This step ensures that the dataset is clean, consistent, and ready for further analysis.

Task	Description
1.	Load the dataset using Pandas.
2.	Display: ○ First 5 rows ○ Last 5 rows ○ Column names ○ Shape of dataset
3.	Check data types of all columns.
4.	Check for missing values in: ○ Selling_Price ○ Present_Price ○ Kms_Driven ○ Fuel_Type
5.	Fill missing values: ○ Selling_Price → mean ○ Present_Price → mean ○ Kms_Driven → mean ○ Fuel_Type → mode
6.	Convert numeric columns to proper numeric type if required: ○ Selling_Price ○ Present_Price ○ Kms_Driven ○ Year
7.	Convert Selling_Price and Kms_Driven into NumPy arrays.

Output:

```
------------------------------------------------------------
  Car_Name  Year  Selling_Price  Present_Price  Kms_Driven Fuel_Type Seller_Type Transmission  Owner
0     ritz  2014           3.35           5.59       27000    Petrol      Dealer       Manual      0
1      sx4  2013           4.75           9.54       43000    Diesel      Dealer       Manual      0
2     ciaz  2017           7.25           9.85        6900    Petrol      Dealer       Manual      0
3  wagon r  2011           2.85           4.15        5200    Petrol      Dealer       Manual      0
4    swift  2014           4.60           6.87       42450    Diesel      Dealer       Manual      0
------------------------------------------------------------

------------------------------------------------------------
    Car_Name  Year  Selling_Price  Present_Price  Kms_Driven Fuel_Type Seller_Type Transmission  Owner
296     city  2016           9.50           11.6       33988    Diesel      Dealer       Manual      0
297     brio  2015           4.00            5.9       60000    Petrol      Dealer       Manual      0
298     city  2009           3.35           11.0       87934    Petrol      Dealer       Manual      0
299     city  2017          11.50           12.5        9000    Diesel      Dealer       Manual      0
300     brio  2016           5.30            5.9        5464    Petrol      Dealer       Manual      0
------------------------------------------------------------

------------------------------------------------------------
Index(['Car_Name', 'Year', 'Selling_Price', 'Present_Price', 'Kms_Driven',
       'Fuel_Type', 'Seller_Type', 'Transmission', 'Owner'],
      dtype='str')
------------------------------------------------------------
(301, 9)
Car_Name             str
Year               int64
Selling_Price    float64
Present_Price    float64
Kms_Driven         int64
Fuel_Type            str
Seller_Type          str
Transmission         str
Owner              int64
dtype: object
Selling_Price    0
Present_Price    0
Kms_Driven       0
Fuel_Type        0
dtype: int64
```

Figure 1.1: Displaying head (), tail () function

Conclusion:

This scenario prepares the dataset by handling missing values and correcting data types.

It ensures data quality and consistency for accurate analysis. Basic statistics provide an initial understanding of selling price distribution. Overall, it builds a strong foundation for further data analysis and visualization.

Scenario 2: Selling Price Trend (Line Graph) (🟢 Easy)

Description:

In this scenario, a subset of the dataset is selected to analyze the trend of selling prices. The first 10 records are extracted using Pandas, focusing on Car Name and Selling Price. The selling price column is converted into a NumPy array for numerical processing. A line graph is plotted using Matplotlib to visualize how selling prices vary across selected entries. The graph includes proper labels, title, and markers for better readability. Finally, the graph is saved as an image file for reporting purposes. This visualization helps in identifying patterns or variations in car selling prices.

Task	Description
1.	Select: ○ Car_Name ○ Selling_Price
2.	Take the first 10 rows only using Pandas.
3.	Convert Selling_Price into a NumPy array.
4.	Plot a line graph using Matplotlib: ○ X-axis → row index (0–9) ○ Y-axis → Selling Price
5.	Add: ○ title ○ x-axis label ○ y-axis label ○ markers
6.	Save the graph with a suitable filename.

Graph:

The line graph shows the variation in selling prices for the first 10 cars in the dataset. The X-axis represents the row index, and the Y-axis represents the selling price. Each point indicates the price of a specific car, highlighting differences of them. The graph shows fluctuations in prices rather than a consistent trend. This indicates that selling price depends on factors like car condition, age, and features.

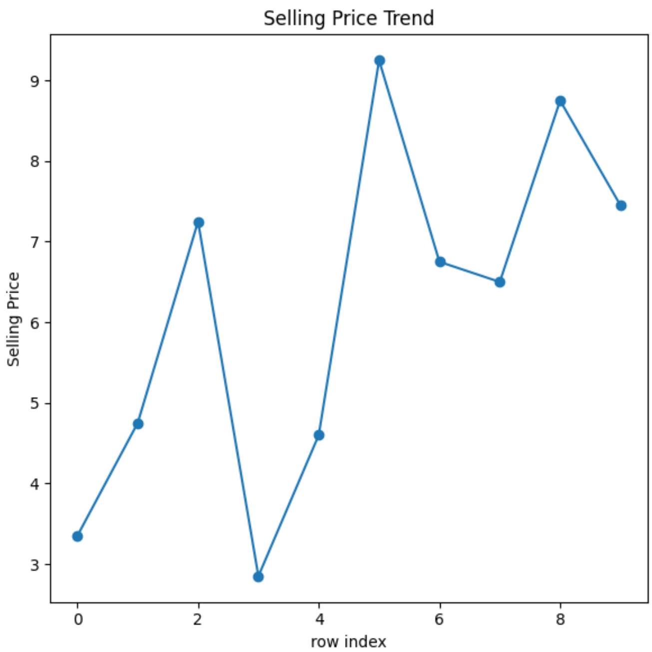

Figure 2.1: Line Graph of Selling Price Trend

Conclusion:

This scenario helps in visualizing the variation in selling prices using a line graph. It makes it easier to identify trends and patterns in the dataset. The use of visualization improves data understanding and interpretation. Overall, it demonstrates how graphical analysis can support decision-making.

Scenario 3: Expensive Cars Analysis (Filtering + Bar) (🟡 Medium)

Description:

In this scenario, the dataset is filtered to analyze cars with a selling price greater than 10. The filtered data is then grouped by fuel type to identify which fuel categories are most common among these high-value cars. The categorical labels and their respective counts are converted into NumPy arrays to facilitate plotting. Finally, a bar chart is generated using Matplotlib to visualize the prevalence of different fuel types in the premium car segment.

Task	Description
1.	Filter cars where: ○ Selling_Price > 10
2.	Group the filtered data by: ○ Fuel_Type
3.	Count number of cars in each fuel type.
4.	Convert: ○ fuel type labels ○ counts into NumPy arrays.
5.	Plot a bar chart using Matplotlib: ○ X-axis → Fuel Type ○ Y-axis → Count of expensive cars
6.	Add: ○ title ○ x-label ○ y-label

Graph:

Filtering for cars priced above 10 shows the fuel preferences in the premium segment. A bar chart reveals that Diesel vehicles heavily dominate these higher-priced sales. This indicates buyers of expensive cars prefer Diesel engines over Petrol options.

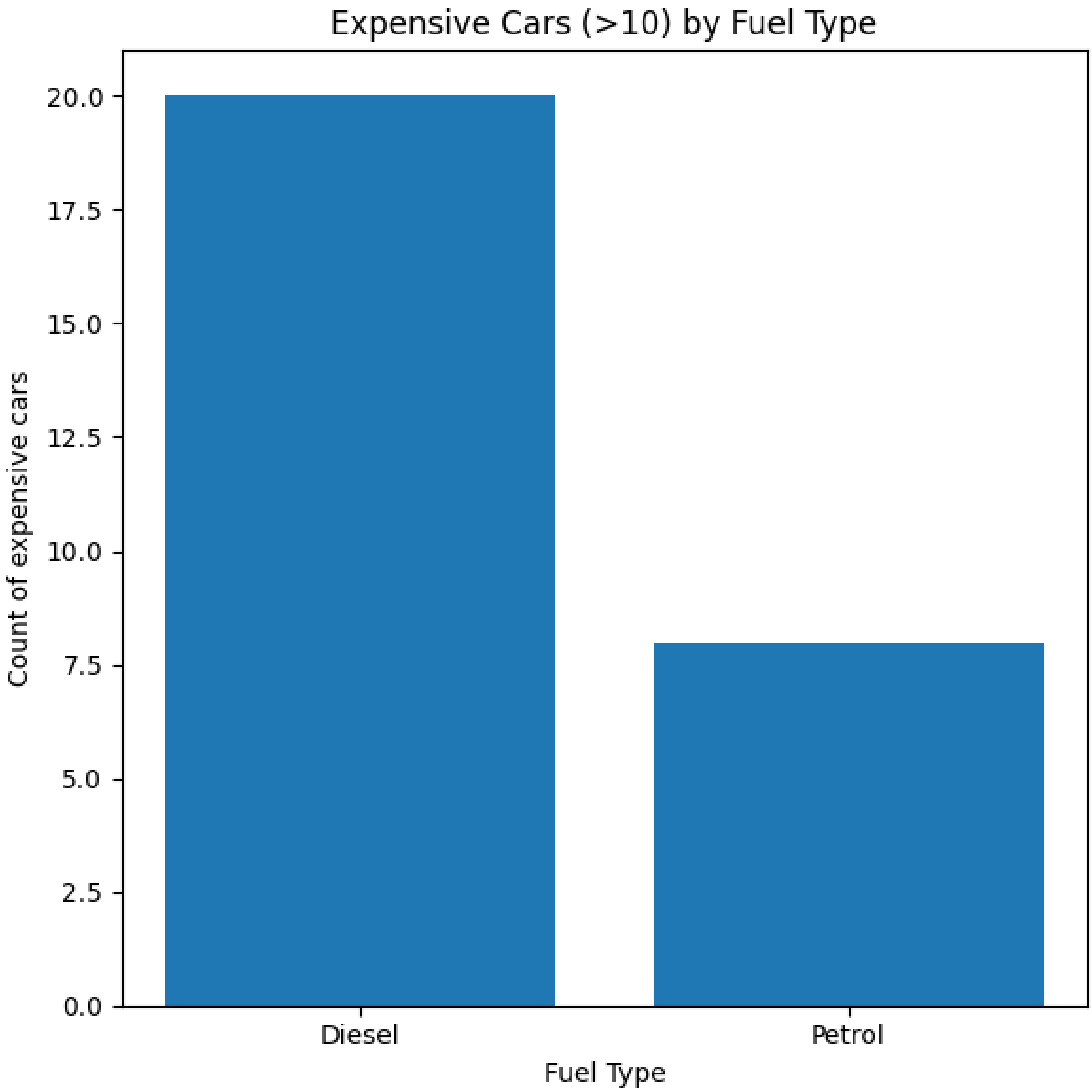

Figure 3.1: Bar Chart of Expencive Cars by Fuel Type

Conclusion:

The analysis reveals the dominant fuel types among more expensive vehicles. By filtering for high-value cars, the bar chart shows that certain fuel types, such as Diesel, are significantly more common in the premium market compared to others. This visualization helps in understanding the relationship between a car's valuation and its required fuel type, providing useful market insights.

Scenario 4: Fuel Type Distribution (Pie Chart) (🟡 Medium)

Description:

This scenario focuses on understanding the overall market composition based on fuel type. The total number of cars in each fuel category is counted using Pandas. The resulting labels and percentage values are prepared and converted into NumPy arrays. A pie chart is plotted using Matplotlib, complete with percentage labels, to provide a clear proportional view of how different fuel types are distributed across the entire dataset.

Task	**Description**
1.	Count the number of cars in each: ○ Fuel_Type
2.	Select all categories or top categories if needed.
3.	Prepare: ○ labels ○ values
4.	Convert values into a NumPy array.
5.	Plot a pie chart using Matplotlib.
6.	Add: ○ percentage labels ○ title

Graph:

A pie chart visualizes the overall market share for different vehicle fuel categories. Petrol vehicles hold the vast majority of the overall dataset at 79.4%. Diesel accounts for 19.9%, leaving CNG as a very small minority at 0.7%.

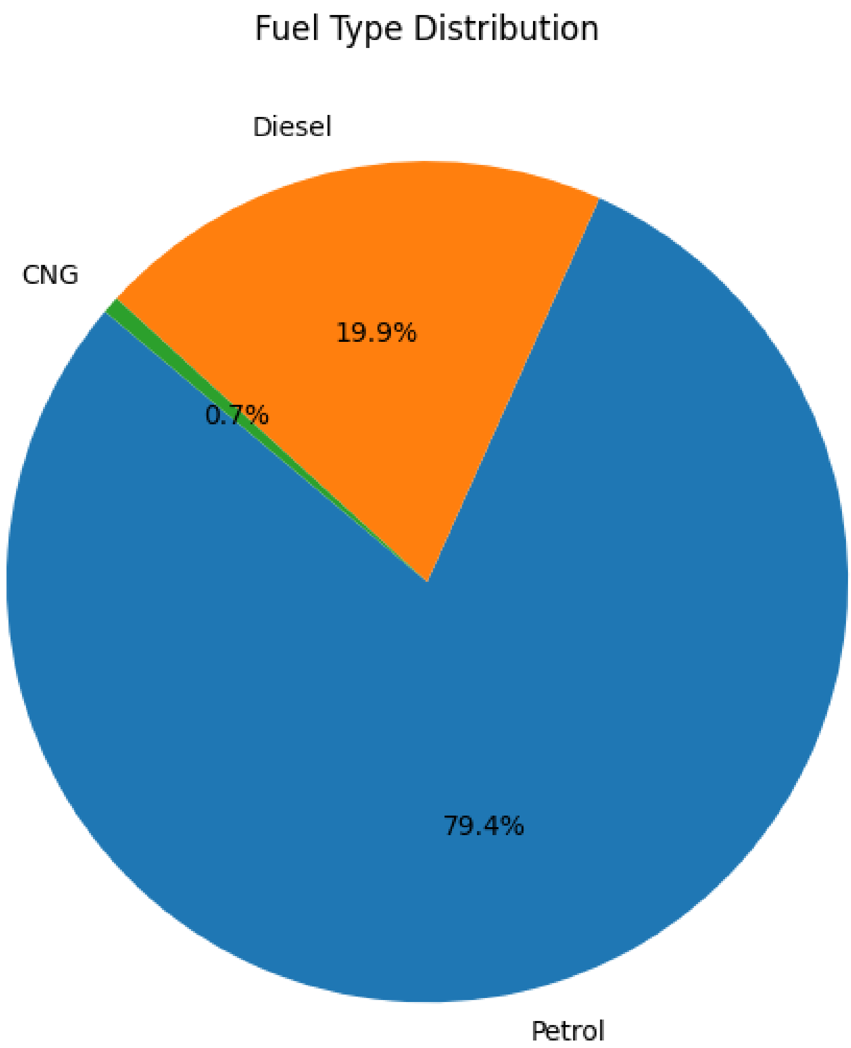

Figure 4.1: Pie Chart of Fuel Type Distribution

Conclusion:

The pie chart effectively highlights the market share of each fuel type. It visually demonstrates that one specific fuel type (typically Petrol) holds the vast majority of the market, while others like Diesel and CNG make up smaller segments. This proportional breakdown is highly valuable for understanding overall consumer choices and inventory distribution.

Scenario 5: Present Price vs Selling Price (Scatter Plot) (🔴 Hard)

Description:

To explore the correlation between a car's initial cost and its resale value, this scenario compares the Present Price against the Selling Price. After removing any missing values, a smaller sample of the first 100 rows is extracted for clearer visualization. The selected columns are converted into NumPy arrays, and a scatter plot is generated. This visual approach helps in determining whether a positive relationship exists between the two pricing metrics.

Task	Description
1.	Select: ○ Present_Price ○ Selling_Price
2.	Remove missing values if any.
3.	Take a smaller sample (for example first 50 or 100 rows) using Panda
4.	Convert both columns into NumPy arrays.
5.	Plot a scatter plot using Matplotlib: ○ X-axis → Present_Price ○ Y-axis → Selling_Price
6.	Add: ○ title ○ x-label ○ y-label
7.	Observe whether there is a positive relationship.

Graph:

A scatter plot compares a vehicle's initial showroom cost to its final resale value. The upward-trending data points confirm a strong positive correlation between both prices. This proves that cars with higher original prices predictably retain higher market values.

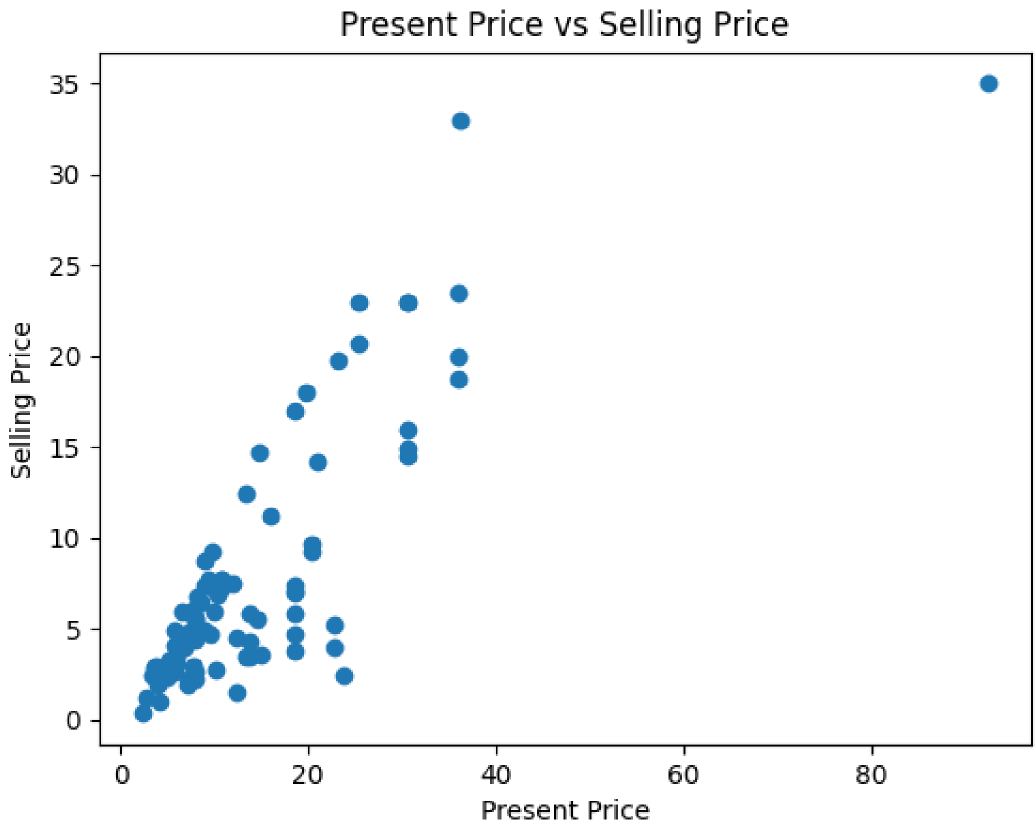

Figure 5.1: Scater plot of Present Price vs Selling Price

Conclusion:

The scatter plot illustrates a distinct positive relationship between present price and selling price. As the present price of a car increases, its selling price generally trends upward as well. This confirms the logical assumption that vehicles with a higher initial showroom valuation retain a proportionally higher resale value in the secondary market.

Scenario 6: Car Age Category Analysis + Bar Chart (🔴 Hard)

Description:

In this scenario, feature engineering is applied to create a new meaningful attribute. Using a custom function in Pandas, cars are categorized based on their manufacturing year into three groups: "New" (2015 and later), "Medium" (2010 to 2014), and "Old" (before 2010). The dataset is then grouped to count the number of cars in each age category. The results are converted into NumPy arrays and visualized via a bar chart to show inventory age distribution.

Task	**Description**
1.	Create a new column using Pandas: Car Age Category ● Year >= 2015 → "New" ● 2010 to 2014 → "Medium" ● < 2010 → "Old"
2.	Count number of cars in each: ○ Car Age Category
3.	Convert category names and counts into NumPy arrays.
4.	Plot a bar chart using Matplotlib: ○ X-axis → Car Age Category ○ Y-axis → Count
5.	Plot a scatter plot using Matplotlib: ○ X-axis → Present_Price ○ Y-axis → Selling_Price
6.	Add title and labels.

Graph:

Vehicles are classified into "New", "Medium", and "Old" based on their manufacturing year. A bar chart demonstrates that newer and moderately aged cars make up most of the inventory. Older cars represent a very small fraction, reflecting typical automotive market turnover trends.

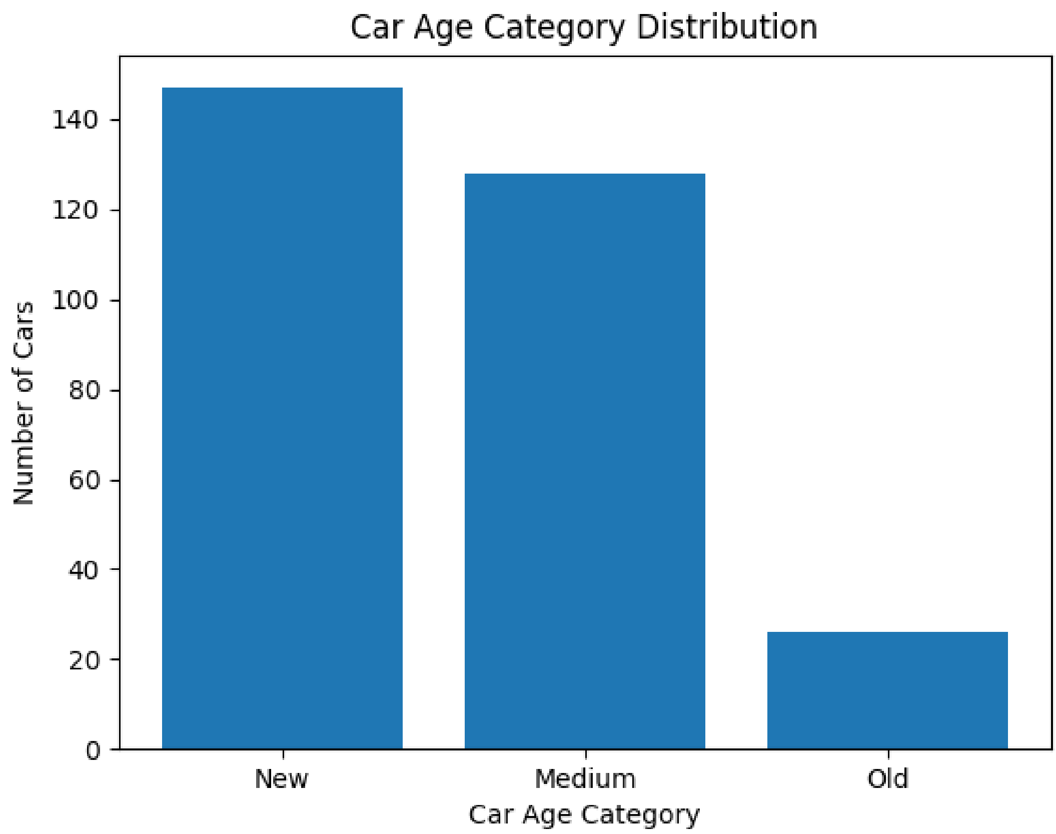

Figure 6.1: Bar Chart of Ca Age Category Distribution

Conclusion:

The bar chart analysis provides a clear view of the age profile of cars in the dataset. It shows that the majority of the vehicles fall into the "New" or "Medium" categories, indicating a highly active market for relatively recent models. Older cars represent a much smaller portion of the data, which aligns with typical vehicle lifespan and market turnover trends.

Scenario 7: Kms Driven Distribution (Histogram) (🔴 Hard)

Description:

In this Scenario, the histogram shows how cars are distributed based on kilometres driven. The X-axis represents the Kms Driven, while the Y-axis shows the number of cars in each range. Most cars are concentrated in the lower to medium mileage range. As mileage increases, the number of cars decreases. This indicates that fewer vehicles have very high mileage.

Task	Description
1.	Select: ○ Kms_Driven
2.	Convert it into a NumPy array.
3.	Plot a histogram using Matplotlib: ○ X-axis → Kms Driven ○ Y-axis → Frequency
4.	Choose suitable number of bins.
5.	Add: ○ title ○ x-label ○ y-label
6.	Save the graph.
7.	Observe whether most cars have lower or higher mileage.

Graph:

The histogram shows the distribution of cars based on kilometers driven. Most cars fall in the lower mileage range, with fewer cars at higher mileage levels. This indicates a right-skewed distribution.

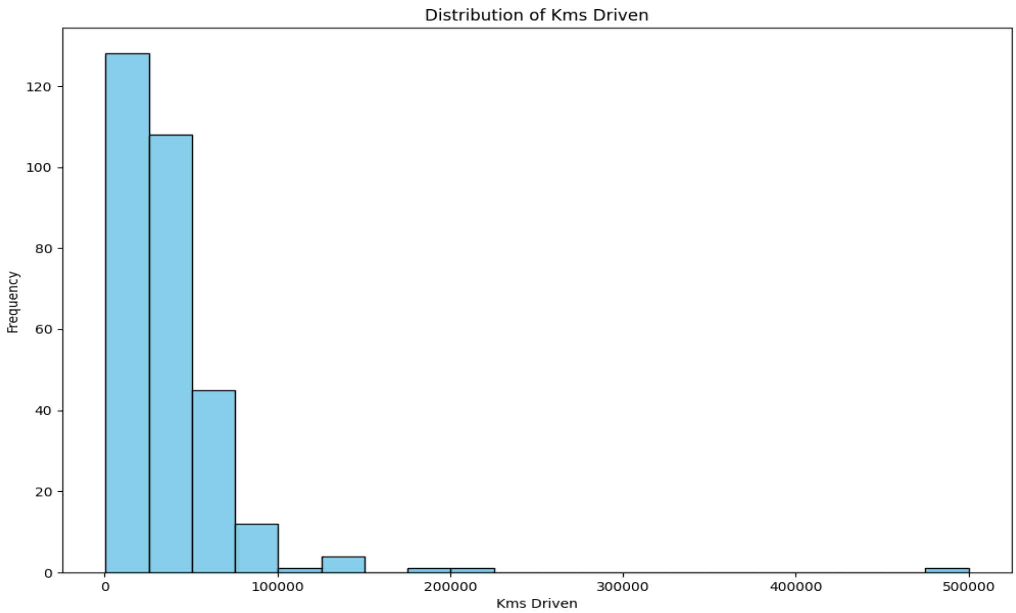

Fig 7.1: Histogram for Distribution of Kms Driven.

Conclusion:

The analysis of the Kms Driven distribution shows that most cars have been driven a relatively low to moderate distance. The frequency decreases as the mileage increases, indicating fewer high-mileage vehicles in the dataset. This suggests that the dataset is dominated by newer or less-used cars. The right-skewed pattern highlights that extreme high values are less common. Such a distribution is useful for understanding vehicle usage trends. Overall, the histogram helps identify that lower mileage cars are more prevalent in the dataset.

Scenario 8: Transmission-wise Selling Price Comparison (🔴 Hard)

Description:

In this Scenario, the bar chart compares the average selling price of cars based on their transmission type. The data is grouped into two categories: manual and automatic. The X-axis represents the transmission types, while the Y-axis shows the average selling price for each category. This visualization helps in understanding how transmission type affects the car's market value. It is often observed that automatic cars have a higher average selling price compared to manual cars. Overall, the chart provides a clear and simple comparison between the two transmission types.

Task	Description
1.	Group data by: ○ Transmission
2.	Calculate: ○ average Selling_Price
3.	Convert transmission labels and average prices into NumPy arrays.
4.	Plot a bar chart using Matplotlib: ○ X-axis → Transmission ○ Y-axis → Average Selling Price
5.	Add title and labels.
6.	Save the graph.

Graph:

The bar chart shows the comparison of average selling prices between manual and automatic cars. Automatic cars generally have a higher average price than manual cars. This indicates that transmission type influences the car’s market value.

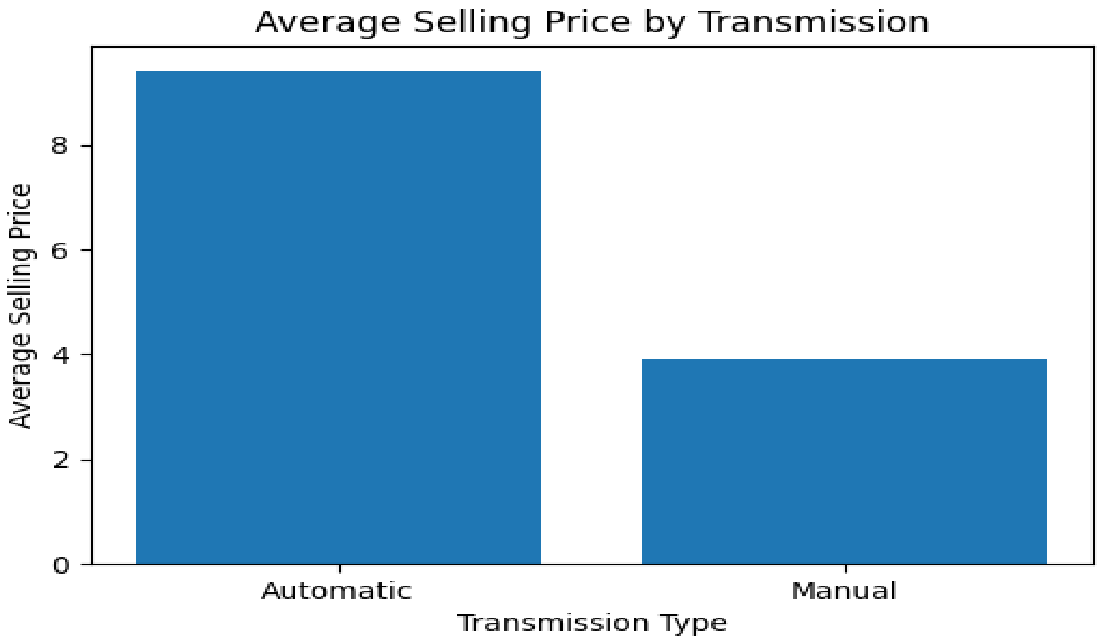

Fig 8.1: Bar Chart for Average Selling Price by Transmission

Conclusion:

The analysis shows that transmission type plays an important role in determining the selling price of cars. Automatic cars generally have a higher average selling price compared to manual cars. This may be due to their convenience, advanced features, and higher demand in the market. Manual cars, on the other hand, are usually more affordable and budget-friendly. The difference in pricing highlights customer preferences and market trends. Overall, transmission type is a key factor influencing car valuation.

Scenario 9: Seller Type Analysis (🔴 Hard)

Description:

This scenario analyzes the distribution of cars based on seller type, such as dealers and individuals. It counts how many cars are sold by each category to understand their contribution to the market. The results are visualized using a bar chart or pie chart for easy comparison. This helps identify which seller type is more dominant. The analysis provides insight into overall selling patterns in the dataset.

Task	Description
1.	Count number of cars by: ○ Seller_Type
2.	Convert results into NumPy arrays.
3.	Plot a bar chart or pie chart using Matplotlib.
4.	Add labels and title.
5.	Save the graph.
6.	Identify which seller type is more common.

Graph :

The graph shows the distribution of cars sold by dealers and individuals. One seller type has a noticeably higher count, indicating it dominates the market. This makes it easy to compare their overall contribution.

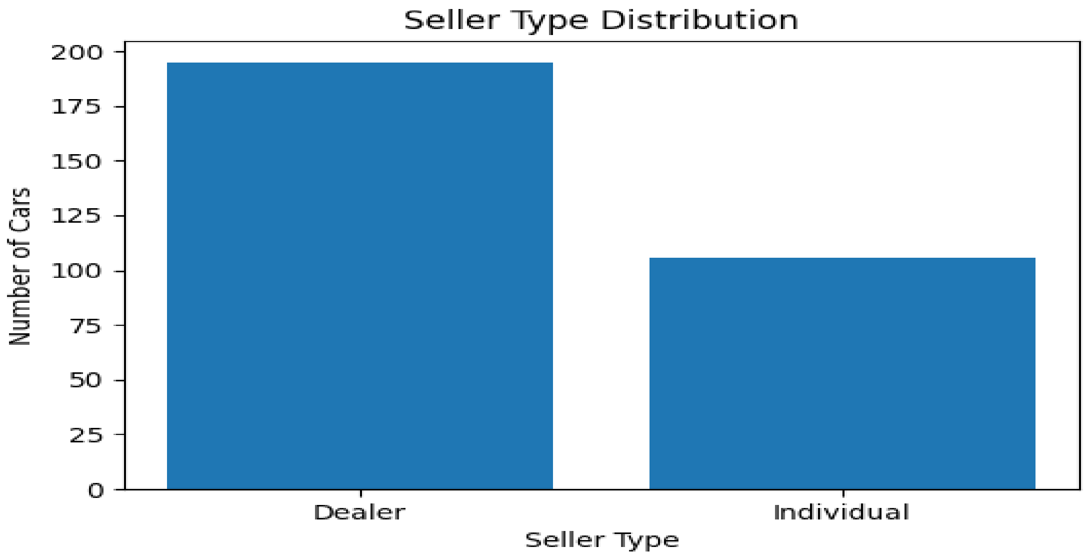

Fig 9.1: Bar Chart for Seller Type Distribution.

Conclusion:

The analysis shows that seller type has a clear impact on the number of cars sold in the dataset. Dealers usually sell a higher number of cars compared to individual sellers. This indicates that dealerships dominate the car market due to their wider reach and resources. Individual sellers contribute a smaller share, often selling fewer vehicles. The difference highlights market structure and selling patterns. Overall, seller type is an important factor in understanding car sales distribution.

Scenario 10: Advanced Analysis + Multiple Graphs (🔴 Hard)

Description:

This scenario performs advanced data analysis by creating a new feature called price difference to measure depreciation. NumPy is used to compute statistical values like average, maximum, and minimum depreciation. Multiple visualizations such as line graphs, bar charts, and histograms help in understanding trends and distributions. The analysis provides deeper insights into pricing patterns, fuel type impact, and car value depreciation.

Task	Description
1.	Create a new column: Price Difference ● Price Difference = Present_Price - Selling_Price This shows how much value the car has depreciated.
2.	Convert Selling_ Price into a NumPy array. ● Use NumPy to calculate price changes between consecutive rows using: ○ np.diff() ● Convert Price Difference column into a NumPy array. ● Find: ○ average depreciation ○ maximum depreciation ○ minimum depreciation
3.	Visualizations □ □ Line Graph ● Plot Selling_Price trend for all cars. □ □ Bar Chart ● Show average Selling_Price by Fuel_Type. □ □ Histogram ● Plot distribution of Selling_Price.

4.	Insights Answer these: ● Which fuel type has the highest average selling price? ● Which transmission type has higher average selling price? ● Are most cars concentrated in lower selling prices or higher selling prices? ● Do older cars tend to have lower selling prices?
5.	Save the graph.
6.	Identify which seller type is more common.

Graph :

1. **Line Graph:**

   The line graph shows how the selling price varies across different cars in the dataset. There are fluctuations, indicating variation in car prices. It highlights that prices are not uniform and depend on multiple factors.

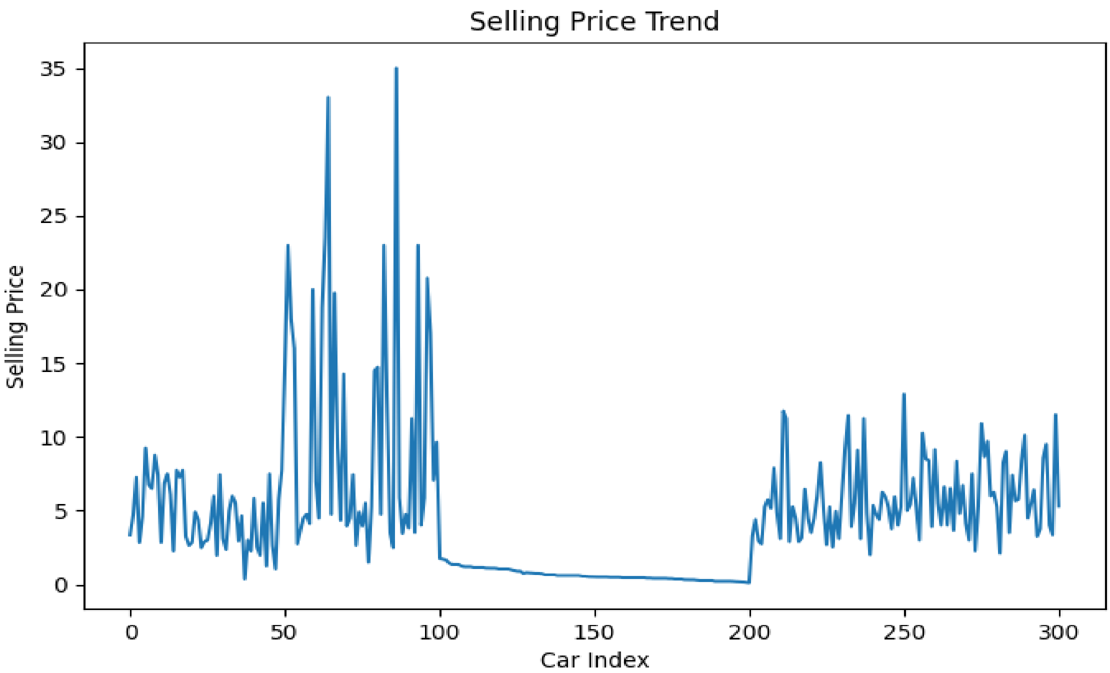

Fig 10.1: Line Graph for Selling Price Trend.

2. **Bar Chart:**

   The bar chart compares the average selling price across different fuel types. One fuel type (usually Petrol/Diesel) shows a higher average price than others. This helps identify which fuel type has greater market value.

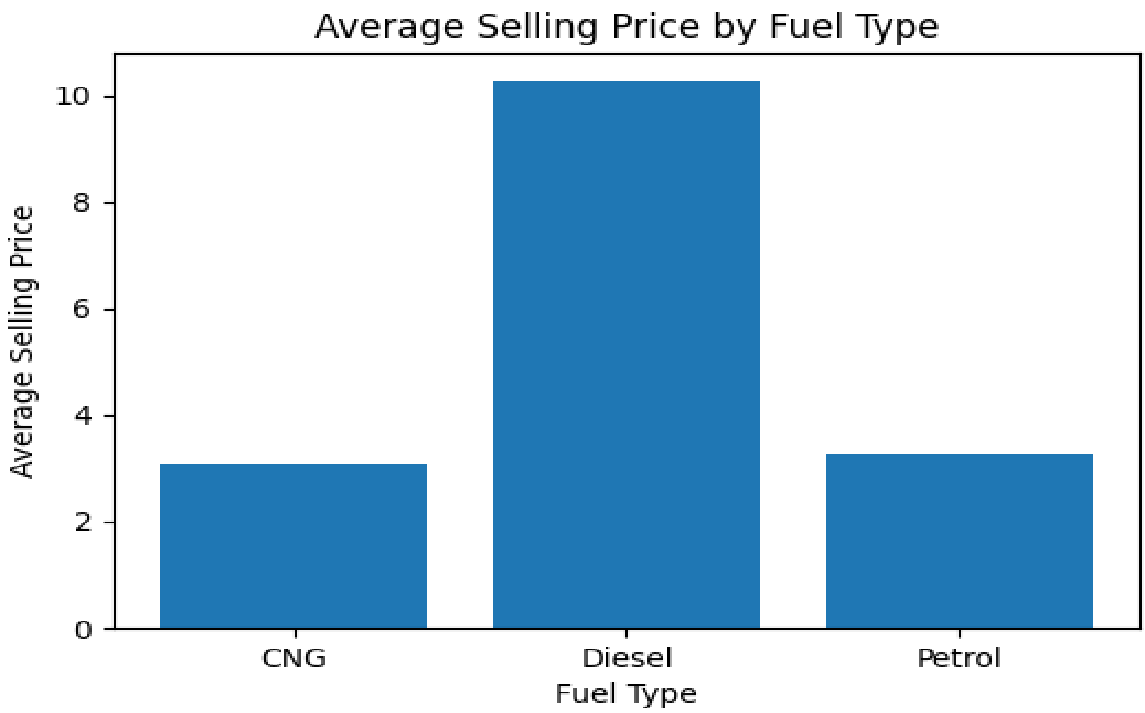

Fig 10.2: Bar Chart for Average Selling Price.

3. **Histogram:**

The histogram shows the distribution of selling prices among cars. Most cars are concentrated in the lower price range, with fewer cars in higher price ranges. This indicates a right-skewed distribution.

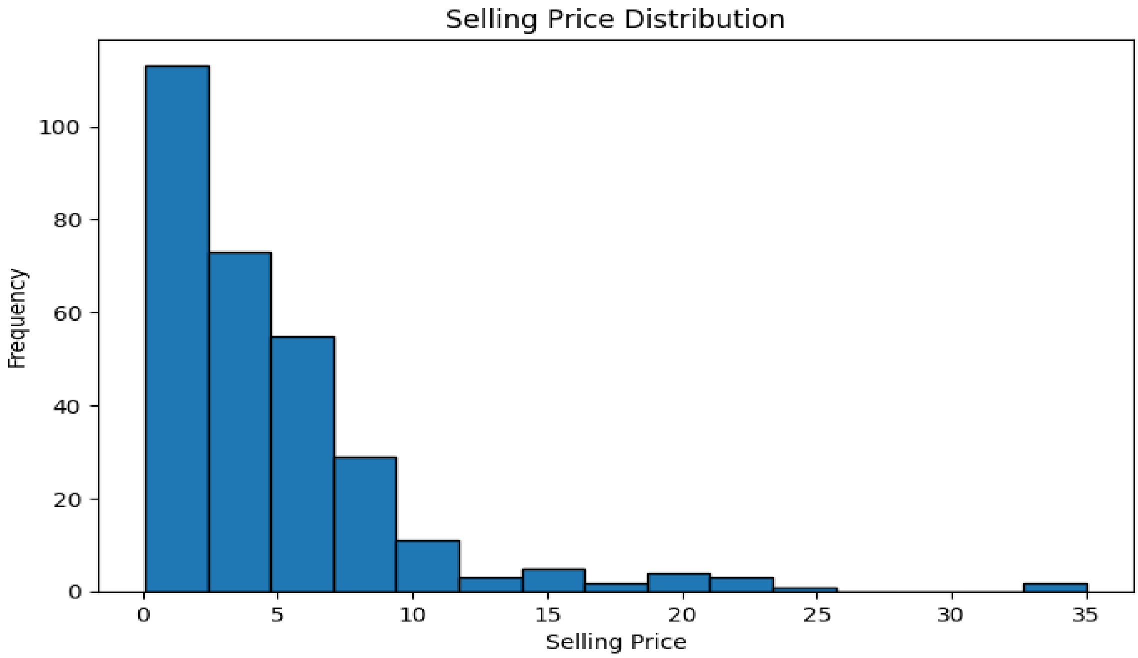

Fig 10.3: Histogram for Selling Price Distribution.

Conclusion:

The advanced analysis provides deeper insights into car pricing and depreciation patterns. The price difference feature shows how much value cars lose over time, indicating depreciation. Most cars are concentrated in the lower selling price range, as seen in the histogram. Fuel type and transmission type both influence the selling price, with certain categories having higher average values. The line graph highlights variations in prices across different cars. Overall, the analysis helps understand market trends and key factors affecting car prices.

Final Conclusion:

The analysis of all 10 scenarios provides a complete understanding of the car dataset from basic to advanced level. It includes data cleaning, feature engineering, and statistical analysis to prepare meaningful insights. Different visualizations such as line graphs, bar charts, pie charts, scatter plots, and histograms help in identifying patterns in the data. The study shows how factors like fuel type, transmission, seller type, and car age affect selling price. Advanced analysis also highlights depreciation and price variation trends. Overall, the project gives a clear and structured view of car market behaviour using data analytics.

What you have learned

This project helped me understand the complete data analysis process step by step, from loading data to generating insights. I learned how to clean and preprocess datasets using Pandas and handle missing values effectively. I gained experience in using NumPy for numerical operations and statistical calculations. I also learned feature engineering by creating new columns like price difference and car age category. Different visualizations such as bar charts, line graphs, scatter plots, pie charts, and histograms improved my data interpretation skills. The project helped me understand how different factors influence car selling prices. Overall, it enhanced my practical knowledge of Python, data analysis, and visualization techniques.